CREATING THE DATABASE SCHEMA

The database must have a table defined (as shown in figure 5.5) for the mapping to work. Spring Data offers a feature to initialize a data source at startup time. By default, you can use a schema.sql file to create a schema and a data.sql file to insert data in the newly created tables. Such files should be placed in the src/main/resources folder.

That is a convenient feature, and it's useful for demos and experiments. However, it's too limited for use in production. As you'll see later in the chapter, it's better to create and evolve relational resources with a more sophisticated tool, like Flyway or Liquibase, which will let you version-control your database. For now we'll use the built-in database initialization mechanism so that we can focus on the data layer implementation first.

NOTE Hibernate, the foundation for Spring Data JPA, offers an interesting feature for automatically generating schemas from the entities defined in Java. Once again, this is convenient for demos and experiments, but please think twice before using it in production.

In your Catalog Service project, add a new schema.sql file in the src/main/resources folder. Then write the SQL instructions to create the `book` table, which will be mapped to the `Book` record in Java.

Listing 5.7 Defining the SQL instructions to create the `book` table

```
DROP TABLE IF EXISTS book;                          ❶
CREATE TABLE book (
  id                BIGSERIAL PRIMARY KEY NOT NULL, ❷
  author            varchar(255) NOT NULL,
  isbn              varchar(255) UNIQUE NOT NULL,   ❸
  price             float8 NOT NULL,
  title             varchar(255) NOT NULL,          ❹
  version           integer NOT NULL                ❺
);
```

❶ Drops the book table if it already exists